

Soumya Ranjan

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Education

IIT Madras, BS in Data Science <i>Aspiring AI/ML Engineer & Data Engineer</i>	Expected Sep 2027 CGPA: 7 / 10
S.E.M. High School, XII Grade (PCM)	Mar 2022

Experience

Growth Cap Partners <i>Software Engineering Intern</i>	Mar 2024 - Apr 2024 On-site
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- Built software solutions improving business process efficiency by 25% across internal operations
- Partnered with cross-functional teams to architect and deploy 3+ technology solutions

Founder & CEO, SoApp <i>NFC/RFID Self-Checkout Startup</i>	Ongoing Nirmaan, IIT Madras
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- Architected NFC/RFID-based self-checkout platform reducing checkout time by 40% with real-time inventory sync
- Engineered end-to-end hardware-software integration with mobile app supporting 50+ retail SKUs
- Patent application filed; Incubation at Nirmaan (IIT Madras Pre-Incubation Cell)

Projects

Smart MCQ Solver	NLP/GenAI, Transformers, Hugging Face, RAG
Developed AI ranking system achieving optimized MAP@3 metric for top-3 MCQ prediction; improved accuracy by 15% via ensemble methods	

- Integrated transformer-based reasoning with retrieval-augmented generation (RAG) for enhanced model performance
- Deployed production-ready Streamlit application supporting real-time inference and user testing

Industrial Equipment Selling Price Prediction	Python, XGBoost, LightGBM, CatBoost
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- Developed tree-ensemble regression pipeline achieving 0.18 RMSLE for heavy-equipment valuation across 45+ features
- Optimized CatBoost, XGBoost, LightGBM models via RandomizedSearchCV, delivering 155% performance gain over baseline
- Built advanced feature encodings (target/frequency) with stratified cross-validation to prevent data leakage

Bank Customer Churn Prediction	Python, XGBoost, Scikit-learn
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- Built classification pipeline achieving 0.85 F1-score; benchmarked 4+ algorithms on imbalanced dataset (30% minority class)
- Implemented class-weighted training and custom threshold optimization to maximize F1-score by 8% over default cutoff
- Built end-to-end ML pipeline with data preprocessing, model selection, and hyperparameter optimization achieving production-ready performance

College Placement Portal	Vue.js, Node.js, Flask, SQLAlchemy, Redis, JWT
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- Architected role-based placement platform supporting 500+ users (students, recruiters, admins) with Vue.js SPA and JWT-based RBAC
- Built async notification system using Celery/Redis, reducing email delivery latency by 60%
- Implemented dual API architecture (REST + GraphQL) enabling flexible client queries and real-time interview scheduling

Technical Skills

Languages: Python, Java, JavaScript, SQL, Bash
ML/Data: Scikit-learn, CatBoost, XGBoost, LightGBM, Random Forest, PyTorch, Pandas, NumPy
NLP & GenAI: Transformers, Hugging Face, RAG, Prompt Engineering, LLM Fine-Tuning, ANN, RNN, DNN
Backend & Web: Flask, SQLAlchemy, REST APIs, GraphQL, JWT, Celery, Redis, Node.js, Vue.js
Databases & Cloud: PostgreSQL, MySQL, AWS, Docker, Linux, Git, CI/CD
Visualisation: Matplotlib, Seaborn, Power BI, Tableau

Leadership & Achievements

Founder & CEO, SoApp NFC/RFID self-checkout startup for retail; incubated at Nirmaan (IIT Madras), patent application under review.

Treasurer & former Society Affairs Head, Shah Maat (IIT Madras BS Chess Club) Managed finances and coordinated events for a 500+ member organization.

Diploma in Programming, IIT Madras Coursework in Application Development, Data Structures & Algorithms, Database Management Systems, and System Commands.

Diploma in Data Science, IIT Madras Coursework in Machine Learning, Generative AI, and Data Science tools.